XCube Platform

Video wall & remote display management
for transportation operations

VIDEO WALL & REMOTE DISPLAY MANAGEMENT

Video walls without the video wall budget.

XCube lets transportation agencies build and drive video walls and remote displays from modular, low-cost device units — often an order of magnitude cheaper than traditional monolithic systems.

~10x	**42 cells**	**Standard / Pro / High Density**	**REST API**
lower cost than monolithic walls	per cube unit, in any grid shape	cube unit types	for every capability

THE PLATFORM

What XCube is

Robust video wall and remote display systems, at a fraction of the cost of traditional monolithic hardware.

XCube is a physical display management platform for TSMO agencies. It is highly reliable and modular, with intelligent real-time monitoring and management, a modern browser-based interface, and a fully featured REST API for external-system integration. It scales through the simple addition of XCube device units, with each deployment driving multiple video walls and remote displays. The central Display Manager deploys on private virtualized environments, bare-metal servers, or commercial cloud.

BUILT FOR THE MANY PLACES OPERATIONS NEED EYES ON GLASS:

- Traffic management centers
- Emergency operations centers
- Maintenance & district offices
- Executive & public lobbies
- Remote and satellite displays
- Partner agencies
- First responders
- Field & shift supervisors

CAPABILITIES

A comprehensive platform, end to end

WALL CONTROL & LAYOUT

Intuitive operator interface

Drive every wall from one browser-based view — drag streams, tours, and web pages onto cells presented with video snapshot overlays, switch layouts, and monitor unit health at a glance. Preview an input with Play before it reaches the wall.

Dynamic layout selection

Switch a wall between dozens of layouts on the fly — any grid from a single 1x1 display up to 42 cells, including deep oblongs such as 8x3 for ribbon and ticker arrangements — no reconfiguration, no downtime.

Wall presets

Save and recall wall layouts — merged cells included — and its stream and tour assignments as named presets.

Single sign-on

Sign in through Microsoft Entra or any OpenID Connect provider, inheriting your organization's MFA and password policy. Provider-managed and local accounts coexist on one deployment, with wall and entity permissions still owned by XCube.

Cell merging

Merge any rectangle of cells, Excel-style, so a single feed spans them — spotlight one camera large while the rest of the grid keeps playing, then unmerge in one action.

Shared control

Several operators can drive one wall from their own browsers: each sees the others' changes, and the history says who moved what.

CONTENT SOURCES

Live video

Assign any configured stream to a cell — a camera, an encoder, or a desktop cast — selected the same way wherever it originates.

XCast desktop sharing

A downloadable Windows app casts a workstation's desktop, any display, or any application window — several at once, each published as an independent stream for wall cell assignment.

Video tours

Cycle a named, ordered sequence of streams through a single cell, advancing after each stream's dwell time and skipping any member that cannot play.

Web pages

Point a cell at any web page — dashboards, weather, incident boards — with an optional automatic reload interval.

CUBES & DISPLAYS

Modular, scalable hardware

Add capacity one XCube unit at a time. Each deployment manages multiple video walls and remote displays from a single platform.

Merged multi-output displays

XCube Pro units merge their physical outputs into one larger display — horizontal, vertical, or quad — with per-connector placement.

SECURITY & MONITORING

Real-time unit monitoring

Every unit reports online / offline state continuously, and can be restarted or rebooted remotely from the browser.

Certificate rotation without downtime

Install a renewed TLS certificate over the REST API — no restart, no dropped viewers — and both the web app and XCast pick it up.

Entity scoping

Scope an account to its entities: the input lists offer only its streams and tours, grouped by entity. A shared wall still plays whatever is on it.

Wall control & sign-in history

Every wall change is recorded — which input went to which cell, what was cleared, moved, or merged — beside a sign-in audit log, per-account session lifetimes, and remote sign-out.

XFLOW INTEGRATION

Best-in-class viewing & control

Paired with the XFlow video management platform, operators watch any stream in the browser at low latency — and pan, tilt and zoom where permitted. Work from a filterable List, the Camera map, or a Personal wall.

Camera map

Every camera on a map, colored live by whether its stream and PTZ are available. Filter by status, search by name, and drag a camera onto the bordering video panel to watch it without leaving the map.

Personal wall

Every operator can build their own wall in the browser, with no cube hardware — fixed grids, or a flex layout that subdivides any cell into quads — assigning streams and tours by drag and drop. Cameras can be controlled in place.

WebRTC with HLS fallback

Cells play over WebRTC, falling back to low-latency HLS where XFlow has ruled it out — and each cell names the protocol carrying it.

INTEROPERABILITY

Supported protocols

SUPPORTED INPUTS

- RTSP / UDP (RTP, RTCP)
- RTSP Multicast
- RTSP / TCP
- RTSPS
- RTMP
- RTMPS
- Multicast TS
- HLS
- DASH

Non-exhaustive. Additional formats can be supported — contact Optima DTS to discuss a specific source. XCube units drive physical displays directly; video is decoded and presented at the cube, not restreamed.

ARCHITECTURE & DEPLOYMENT

Built to scale, deployed your way

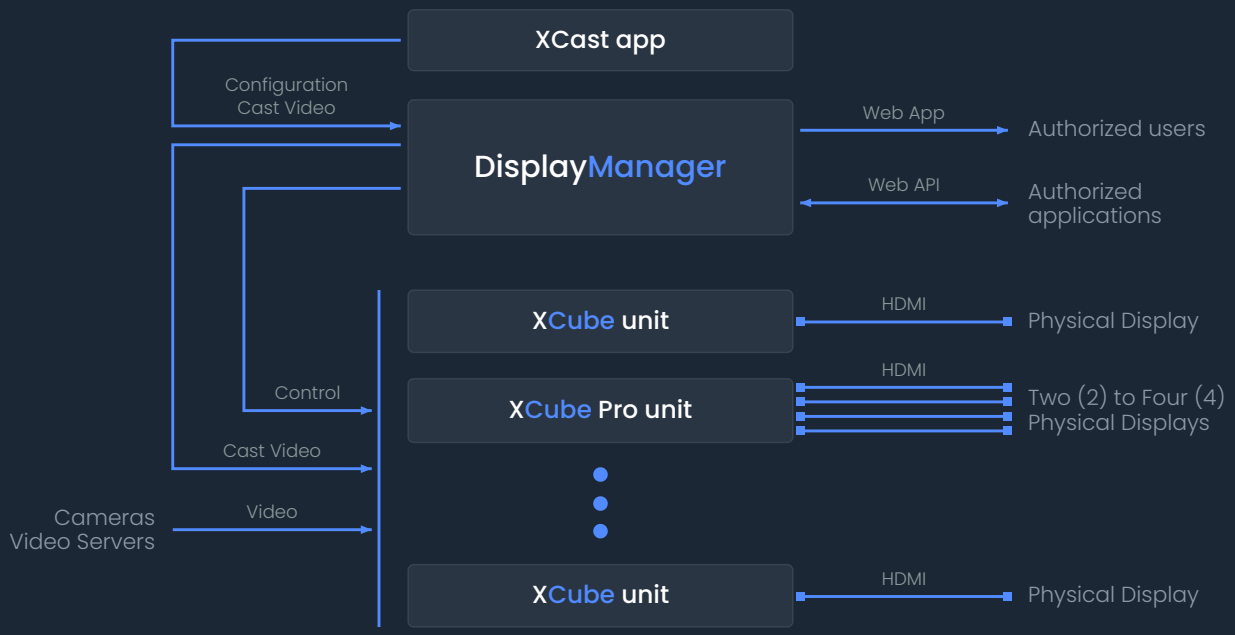
XCube Unit

The device firmware — Go and C/C++ — delivers high-quality video playback at low latency and resource cost.

XCube Display Manager

The central component: control, monitoring, and configuration for the entire platform.

Scale by adding XCube device units. Deploy the Display Manager on private virtualized environments, bare-metal servers, or commercial cloud.



Platform data architecture

Deployed in transportation operations centers to drive video walls and remote displays at a fraction of traditional cost.

Bring XCube to your operation.

Additional source formats can be supported. Contact Optima DTS to discuss your deployment or arrange a demonstration.

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